



JAIPUR NATIONAL UNIVERSITY

Semester/Year End (Main/Back) Examinations, May-2025

Name of School:	School of Engineering and Technology
Name of Programme / Course:	B.Tech. (Computer Science Engineering) / B.Tech. CSE (Hons.) in Cyber Security / B.Tech. CSE (Hons.) in Artificial Intelligence and Machine Learning
Semester/Year:	Semester-VI
Paper Code:	BTECSCO601T22
Title of Paper:	Computer Networks
Time: 3 Hrs.	Max. Marks: 60

INSTRUCTIONS / NOTES FOR THE EXAMINEE

This question paper consists of two sections, Section-A and Section-B. Section-A consists of seven questions and Section-B consists of six questions. Attempt any five questions from Section-A and any four questions from Section-B. Marks allotted are mentioned against each question.

Question Number	SECTION	Marks Allotted
Q.1	Define Computer Network. Explain the various types of computer networks with examples.	4
Q.2	Compare and contrast OSI and TCP/IP models.	4
Q.3	Explain the functions of different layers in the OSI model.	4
Q.4	What is IP addressing? Differentiate between IPv4 and IPv6.	4
Q.5	Explain the concept of MAC addressing and its role in data communication.	4
Q.6	Define congestion in networks. Mention any two congestion control techniques.	4
Q.7	Differentiate between circuit switching and packet switching with suitable examples.	4

SECTION-B

- Q.1** What is routing? Explain distance vector routing and link state routing with examples. **10**
- Q.2** Describe various error detection techniques like Parity Check, CRC, and Checksum. **10**
- Q.3** What is HTTP? Explain the working of HTTP protocol and its request/response cycle. **10**
- Q.4** Describe the working of TCP three-way handshake protocol for establishing a connection. **10**
- Q.5** Explain DNS. How does name resolution work in the DNS hierarchy? **10**
- Q.6** Discuss in detail about Subnetting and Supernetting with suitable examples and diagrams. **10**